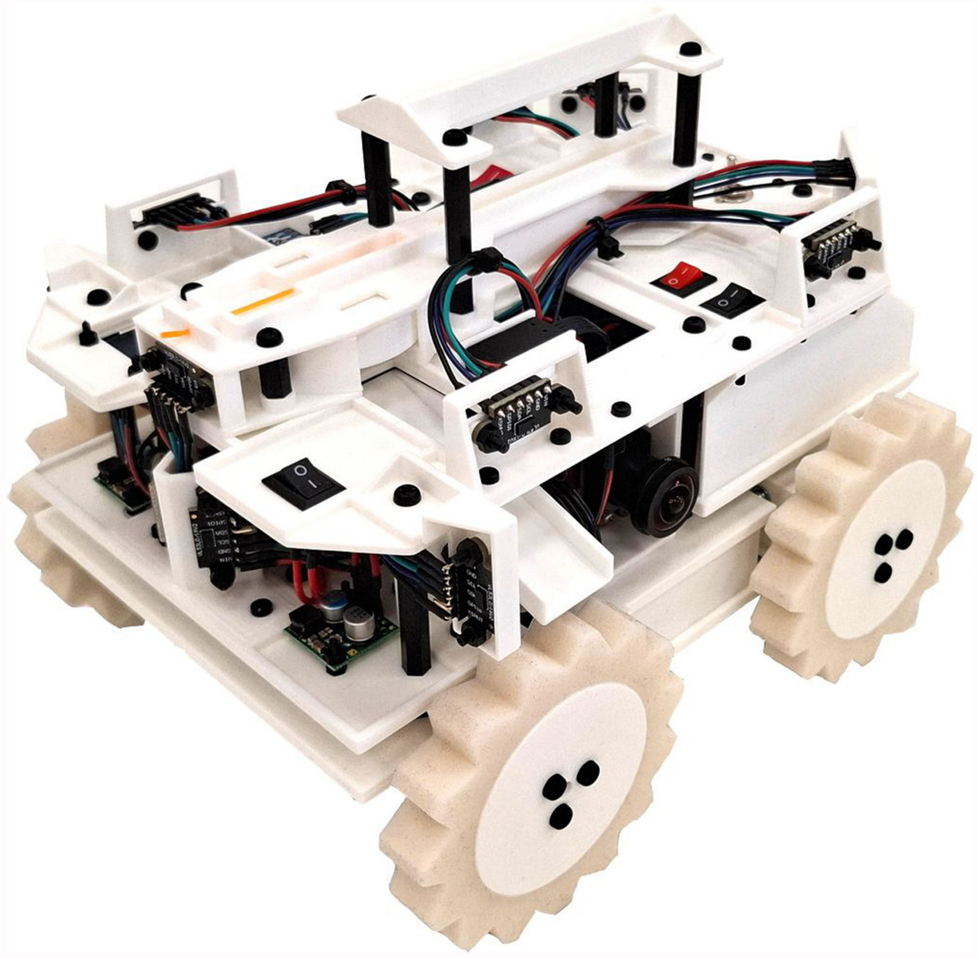
LIGHTNING

Hungarian Robotics Team

RoboCup Junior 2026 Rescue Maze



Team members

Jazmin Szilagyi (captain)
Zsofia Panna Cs. Kiss (co-captain)
Nikolett Szel

Mentor

Csaba Aban

Teamwork

all team members work together on all improvements, but there are priorities:

Zsofia

Electronics
Soldering, wiring
Power supply design
Robot structure
CAD design and assembly
Programming
Program Structure

Jazmin

Programming
Special Cases or Situations
Robot structure
3D printing
Electronics
Electrical circuit planning
Soldering, wiring

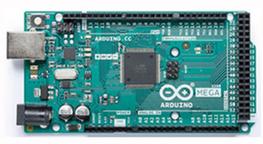
Nikolett

Programming
Program Structure, Sensor Integration
Robot structure
Material selection, assembly
CAD design and assembly
Electronics
Power supply design

Robot hardware

OLED Display (1)
to follow the robot's process

QTRXL Infra Sensor (1)
silver tile detection

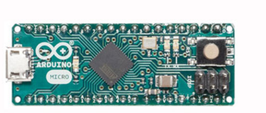


Arduino Mega (1)
main control unit

DC motor 12V (4)
moving the robot

Color Sensor (1)
tile color detection

Voltage Regulator (2)
12,6 V to 5V for the servo
8,4 V to 5V for the OpenMV

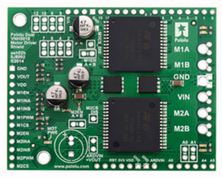


Arduino Micro (1)
to connect the colour sensor
prevent I2C address match
communicate with the Mega

Metal Gear Servo (1)
rescue kit dropper

TOF Sensor (11)
measuring distance
(wall and obstacles)
for navigation and mapping

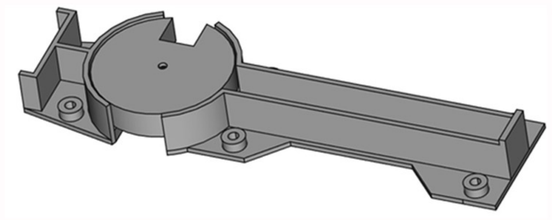
OpenMV H7 Cam (2)
victim detection and
identification



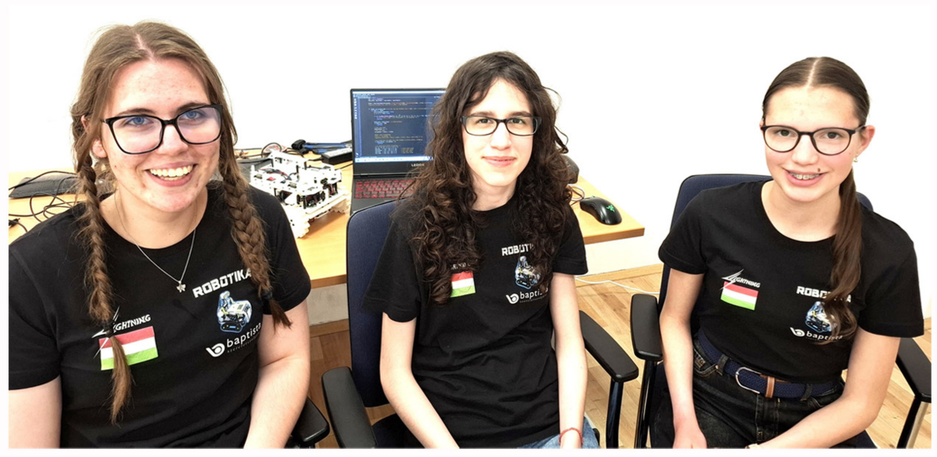
Dual Motor Driver (1)
control of motors

Lipo battery (2)
power supply
3S (12,6 V) motors
2S (8,4 V) microcontrollers

MPU6050 Module (1)
more precise turning
angles, detection of
ramps and stairs



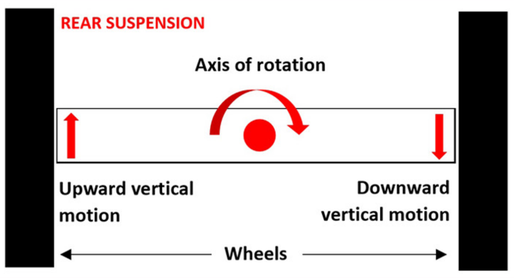
Dual-side rescue kit dropper.



Zsofia Jazmin Nikolett

Interesting hardware solutions

Kinematically Balanced Rear Suspension for Mobile Robots
The robot's rear suspension features a pivoting mechanism along a longitudinal axis. The rear crossbeam is mounted on a central shaft aligned with the robot's longitudinal axis, allowing it to freely rotate. This configuration enables the rear wheels to move in opposite vertical directions—when one wheel is lifted by an obstacle, the other is lowered. This articulation improves terrain adaptability, reduces body tilt, and enhances the robot's ability to traverse and maneuver over uneven terrain.



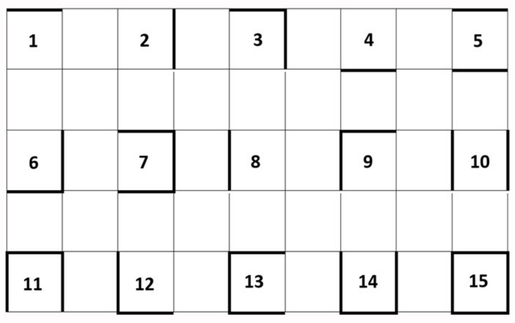
Optimized Tread Design for 2 cm Speed Bump Traversal
The wheel's tread design has been engineered to ensure smooth and efficient passage over 2 cm speed bumps, reducing mechanical stress and maintaining continuous traction.

Interesting electronic solution

Dual Power Supply Architecture for Enhanced System Reliability
The robot is equipped with two separate batteries to ensure stable and reliable power distribution. One battery exclusively supplies the drive system (motors), while the other powers the control units and sensors. This split power supply architecture enhances system reliability and reduces the risk of interference affecting the control system's performance.

Interesting algorithms

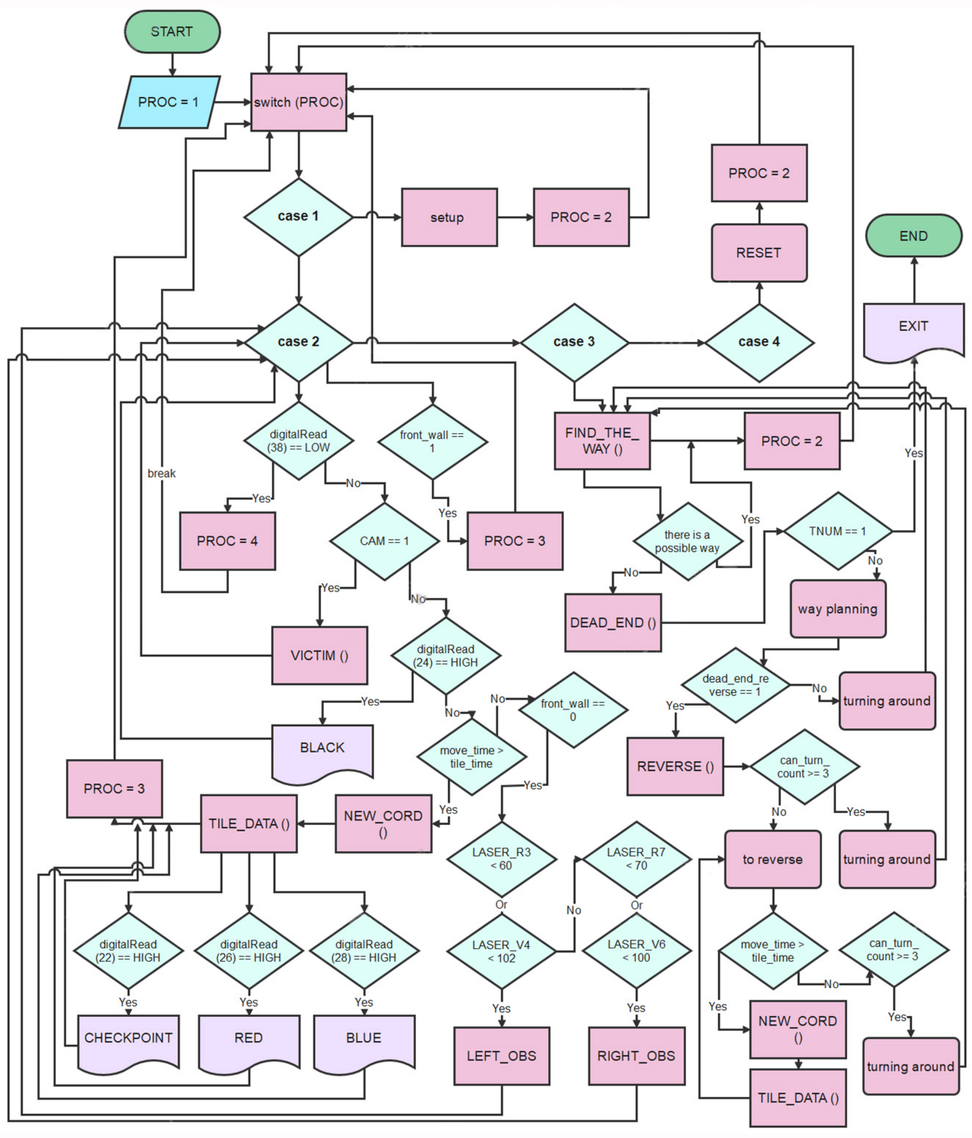
The Arduino-based control program stores data for each tile in an array, where wall configurations detected by distance sensors are encoded as numerical values ranging from 1 to 14. With sensors placed at the front and on both sides, the robot can identify the layout of walls surrounding the current tile. For visited tiles, the encoded value can be at most 14. In dead-end situations, this encoded sensor data supports the path-planning algorithm by enabling the search for alternative routes. In such cases, the robot searches for the nearest accessible tile from which it can move to an adjacent tile that has not yet been visited and is not blocked by a wall.



Robot software

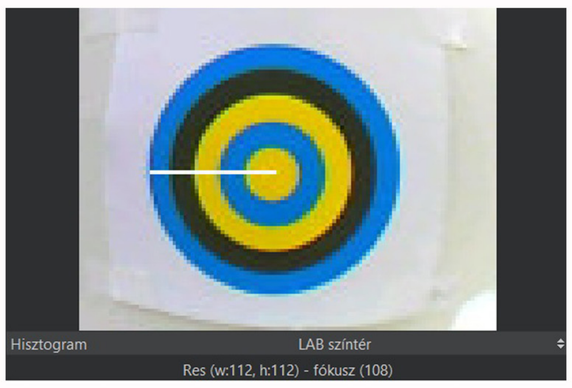
Arduino C, MicroPython

Main Program Structure




```
boolean CHECK_WALL(byte x, byte y, byte d)
{
  boolean w = 0;
  switch(d)
  {
    case 1:
    if (MAP[x][y][1] == 1 || MAP[x][y][1] == 3 || MAP[x][y][1] == 5 || MAP[x][y][1] == 7 || MAP[x][y][1] == 9 || MAP[x][y][1] == 11 || MAP[x][y][1] == 13)
    {w = 1;}
    break;
    case 2:
    if (MAP[x][y][1] == 2 || MAP[x][y][1] == 3 || MAP[x][y][1] == 6 || MAP[x][y][1] == 7 || MAP[x][y][1] == 10 || MAP[x][y][1] == 11 || MAP[x][y][1] == 14)
    {w = 1;}
    break;
    case 3:
    if (MAP[x][y][1] == 4 || MAP[x][y][1] == 5 || MAP[x][y][1] == 6 || MAP[x][y][1] == 7 || MAP[x][y][1] == 12 || MAP[x][y][1] == 13 || MAP[x][y][1] == 14)
    {w = 1;}
    break;
    case 4:
    if (MAP[x][y][1] == 8 || MAP[x][y][1] == 9 || MAP[x][y][1] == 10 || MAP[x][y][1] == 11 || MAP[x][y][1] == 12 || MAP[x][y][1] == 13 || MAP[x][y][1] == 14)
    {w = 1;}
    break;
  }
  return w;
}
```

RCJ2025 Salvador



Gradient- and ring-width-based cognitive casualty analysis.

International RCJ results

Jazmin
EURCJ 2019 (Hannover)
Rescue Line Entry V.
EURCJ 2022 (Guimaraes)
Rescue Maze Entry III.
EURCJ 2023 (Varazdin)
Rescue Line II.
RCJ 2023 (Bordeaux)
Rescue Line Individual Team II.
EURCJ 2024 (Hannover)
Rescue Line II.
RCJ 2024 (Eindhoven)
Rescue Line
Outstanding Engineering Process
EURCJ 2025 (Bari)
Rescue Maze Individual Team II.
RCJ 2025 (Salvador)
Rescue Maze Individual Team II.

Zsofia
EURCJ 2022 (Guimaraes)
Rescue Maze I.
RCJ 2022 (Bangkok)
Rescue Maze SuperTeams I.
Rescue Maze Individual Team II.
EURCJ 2023 (Varazdin)
Rescue Line II.
RCJ 2023 (Bordeaux)
Rescue Line Individual Team II.
EURCJ 2024 (Hannover)
Rescue Line II.
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Nikolett
EURCJ 2022 (Guimaraes)
Rescue Maze Entry III.
EURCJ 2023 (Varazdin)
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